

1. recall that molten compounds of metals with non-metals conduct electricity and that this is evidence that they are made up of charged particles called ions;
2. recall that an ion is an atom (or group of atoms) that has gained or lost electrons and so has an overall charge;
3. account for the charge on the ions of group 1 and group 7 elements by comparing the number and arrangement of the electrons in the atoms and ions of these elements;
4. **work out the formulae of ionic compounds given the charges on the ions;**
5. **work out the charge on one ion given the formula of a salt and the charge on the other ion;**
6. recall that compounds of group 1 metals and group 7 elements are ionic;
7. understand that solid ionic compounds form crystals because the ions are arranged in a regular lattice;
8. describe what happens to the ions when an ionic crystal melts or dissolves in water;
9. **explain that ionic compounds conduct electricity when molten or when dissolved in water because the ions are charged and they are able to move around independently in the liquid.**

Chemistry is fundamental to an understanding of the scale and significance of human impacts on the natural environment. Knowledge of natural processes makes it possible to appreciate the environmental consequences of agriculture and the polluting effects of extracting and processing minerals.

The module uses environmental contexts to introduce theories of structure and bonding. The first topic explains the characteristics of covalent bonding, ionic bonding and intermolecular forces in the context of the chemicals found in the atmosphere and hydrosphere.

The second topic uses chemicals in the Earth's crust, such as silicon dioxide, to demonstrate and describe the properties of giant structures with strong covalent bonding. The third topic shows that the natural environment is not static but that elements move between the spheres. The study of natural cycles features the nature of some chemicals in the biosphere such as proteins. The final topic covers the distribution, structure and properties of metals through a study of their extraction from ores. This includes the use of relative atomic masses to give a quantitative interpretation of chemical formulae.

Topics

C5.1 What types of chemicals make up the atmosphere and hydrosphere?

The structure and properties of chemicals found in the atmosphere and hydrosphere.

C5.2 What types of chemicals make up the Earth's lithosphere?

Relating the properties of chemicals to their giant structure using examples found in the Earth's lithosphere.

C5.3 Which chemicals make up the biosphere?

Composition of chemicals found in the biosphere and the natural cycles of elements between the spheres.

C5.4 How can we extract useful metals from minerals?

Relating the structure and properties of metals to suitable methods of extraction.

Using ionic theory to explain electrolysis. Discussing issues relating to metal extraction and recycling.

ICT Opportunities

This module offers opportunities for illustrating the use of ICT in science. For example, modelling molecules and giant structures to explain properties.

Use of ICT in teaching and learning can include:

- animations to show the movement of molecules in a gas over a range of temperatures;
- modelling software to show the shapes of molecules and illustrate giant structures;